



Student name: _____

For each question, put the code for the correct answer in capital letters: A, B, C, and D in the following table beneath the question number.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40

Choose the correct answer for each of the following questions:

Question (1):

A random process $X(t)$ is defined by

$$X(t) = K \cos wt \quad t \geq 0$$

where w is a constant and K is uniformly distributed between 0 and 2.

(1) The expected value of $X(t)$ is

- | | | | |
|-----------------|---------------|-----------------|-----------------|
| (A) $2 \cos wt$ | (B) $\cos wt$ | (C) $3 \cos wt$ | (D) $4 \cos wt$ |
|-----------------|---------------|-----------------|-----------------|

(2) The autocorrelation function of $X(t)$ is

- | | | | |
|--|--|--|--|
| (A) $\left(\frac{4}{3}\right) \cos(wt) \cos(ws)$ | (B) $\left(\frac{1}{3}\right) \cos(wt) \cos(ws)$ | (C) $\left(\frac{2}{3}\right) \cos(wt) \cos(ws)$ | (D) $\left(\frac{1}{2}\right) \cos(wt) \cos(ws)$ |
|--|--|--|--|

(3) The autocovariance function of $X(t)$ is

- | | | | |
|--|--|--|--|
| (A) $\left(\frac{1}{2}\right) \cos(wt) \cos(ws)$ | (B) $\left(\frac{4}{3}\right) \cos(wt) \cos(ws)$ | (C) $\left(\frac{2}{3}\right) \cos(wt) \cos(ws)$ | (D) $\left(\frac{1}{3}\right) \cos(wt) \cos(ws)$ |
|--|--|--|--|

Question (2):

Consider the following social mobility problem. Studies indicate that people in a society can be classified as belonging to the upper class (state 1), middle class (state 2), and lower class (state 3). Membership in any class is inherited in the following probabilistic manner. Given that a person is raised in an upper-class family, he will have an upper-class family with probability 0.45, a middle-class family with probability 0.48, and a lower-class family with probability 0.07. Given that a person is raised in a middle-class family, he will have an upper-class family with probability 0.05, a middle-class family with probability 0.70, and a lower-class family with probability 0.25. Finally, given that a person is raised in a lower-class family, he will have an upper-class family with probability 0.01, a middle-class family with probability 0.50, and a lower-class family with probability 0.49. Determine the following:

(4) The limiting-state probabilities are

- | | | | |
|------------------------|-----------------------|------------------------|------------------------|
| (A) [0.02, 0.57, 0.41] | (B) [0.25, 0.6, 0.15] | (C) [0.06, 0.62, 0.32] | (D) [0.06, 0.55, 0.39] |
|------------------------|-----------------------|------------------------|------------------------|

(5) Given that the process is currently in state 3, the probability that it will be in state 1 at the end of the second transition is

- | | | | |
|------------|------------|------------|------------|
| (A) 0.0199 | (B) 0.0344 | (C) 0.0266 | (D) 0.0577 |
|------------|------------|------------|------------|

Question (3):

Two jointly stationary random processes $X(t)$ and $Y(t)$ have the crosscorrelation function given by:

$$R_{XY}(\tau) = 2e^{-2\tau} \quad \tau \geq 0$$

(6) The cross-power spectral density $S_{XY}(\omega)$ is

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| (A) $2/(2 + j\omega)$ | (B) $1/(1 + j\omega)$ | (C) $4/(2 + j\omega)$ | (D) $2/(4 + j\omega)$ |
|-----------------------|-----------------------|-----------------------|-----------------------|

(7) The cross-power spectral density $S_{YX}(\omega)$ is

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| (A) $1/(1 - j\omega)$ | (B) $4/(2 - j\omega)$ | (C) $2/(2 - j\omega)$ | (D) $2/(4 - j\omega)$ |
|-----------------------|-----------------------|-----------------------|-----------------------|

Question (4):

Two lightbulbs, labeled L_1 and L_2 , have exponentially distributed lifetimes. If the two lifetimes of the two bulbs are independent and the mean lifetime of bulb L_1 is 700 hours, while the mean lifetime of bulb L_2 is 400 hours.

(8) The mean time to a bulb failure is

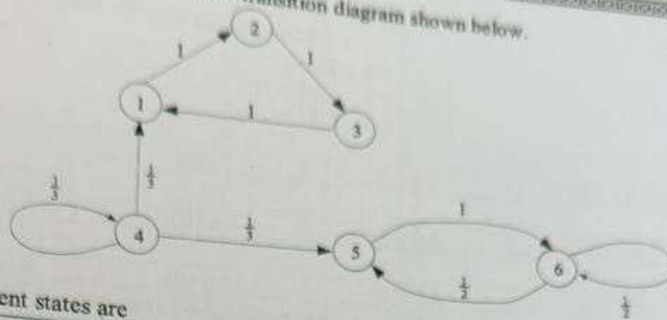
- | | | | |
|----------|---------|---------|---------|
| (A) 1100 | (B) 255 | (C) 300 | (D) 143 |
|----------|---------|---------|---------|

(9) The probability that bulb L_1 fails before bulb L_2 is

- | | | | |
|-----------|-----------|-----------|-----------|
| (A) 0.286 | (B) 0.471 | (C) 0.495 | (D) 0.364 |
|-----------|-----------|-----------|-----------|

Question (5):

Consider a Markov chain with the state-transition diagram shown below.



(10) The recurrent states are

- (A) {1, 3, 4, 5, 6} (B) {1, 2, 3, 4, 5} (C) {1, 2, 4, 5, 6} (D) {1, 2, 3, 5, 6}

(11) The transient states are

- (A) {4} (B) {6} (C) {5} (D) {1, 2, 3}

(12) The number of chains of recurrent states are

- (A) 3 (B) 1 (C) 2 (D) 0

Concerning the transition probability matrix,

(13) The element P_{11} is

- (A) 0.25 (B) 1 (C) 0 (D) 0.5

(14) The element P_{14} is

- (A) 0 (B) 0.5 (C) 1 (D) 0.25

(15) The element P_{32} is

- (A) 0.5 (B) 1 (C) 0.25 (D) 0

(16) The element P_{41} is

- (A) 1 (B) 0.33 (C) 0.25 (D) 0.5

Question (6):

(17) If the arrival rate of a Poisson process is independent of time, the Poisson process is called

- (A) compound (B) homogeneous (C) filtered (D) nonhomogeneous

Question (7):

A bag contains 3 red balls, 6 green balls, and 2 blue balls. Jack plays a game in which he bets \$1 to draw a ball from the bag. If he draws a green ball, he wins \$1 dollar; otherwise he loses a dollar. Assume that the balls are drawn with replacement and that Jack starts the game with \$50 with the hope of reaching \$100 before going bankrupt.

(18) The probability that he will succeed is

- (A) 0.99989 (B) 0.00011 (C) 0.54545 (D) 0.45455

Question (8):

Ten pieces of candy are given out at random to a group that consists of 8 boys, 7 girls, and 5 adults. If anyone can get more than one piece of candy, then:

(19) The probability that 4 pieces go to the girls and 2 go to the adults is

- (A) 0.0976 (B) 0.0679 (C) 0.0756 (D) 0.0473

(20) The probability that 5 pieces go to the boys is

- (A) 0.20066 (B) 0.30077 (C) 0.10044 (D) 0.50088

Question (9):

Consider the following transition probability matrix:

$$P = \begin{bmatrix} 0.4 & 0.3 & P_{13} \\ P_{21} & 0.4 & 0.3 \\ 0.3 & P_{32} & 0.4 \end{bmatrix}$$

(21) The element P_{13} is

(A) 0.25	(B) 0	(C) 0.3	(D) 0.5
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(22) The element P_{21} is

(A) 1	(B) 0.3	(C) 0.5	(D) 0.25
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(23) The element P_{32} is

(A) 0	(B) 0.25	(C) 0.3	(D) 0.5
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(24) The associated limiting-state probabilities are

(A) $[1/3, 1/3, 1/3]$	(B) $[1/3, 0, 2/3]$	(C) $[1/4, 1/2, 1/4]$	(D) $[1/4, 1/4, 1/2]$
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Question (10):

Two random variables X and Y have the joint PDF given by

$$f_{XY}(x, y) = \begin{cases} k(1-x^2y) & 0 \leq x \leq 1; 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(25) The value of the constant k that makes $f_{XY}(x, y)$ a true joint PDF is

(A) 3.5	(B) 1.2	(C) 2.3	(D) 1.5
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(26) The marginal PDF of X is

(A) $k\{1 - 0.25x^3\}$	(B) $k\{1 - 0.25x^2\}$	(C) $k\{1 - 0.5x^2\}$	(D) $k\{1 - 0.5x^3\}$
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(27) The marginal PDF of Y is

(A) $k\{1 - 0.25y^2\}$	(B) $k\{1 - 0.33y^2\}$	(C) $k\{1 - 0.25y\}$	(D) $k\{1 - 0.33y\}$
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(28) The conditional PDF of X given Y , $f_{X|Y}(x|y)$, is

(A) $\frac{3}{3-y}\{1 - xy^2\}$	(B) $\frac{3}{3-y}\{1 - x^2y\}$	(C) $\frac{2}{2-y}\{1 - x^2y\}$	(D) $\frac{2}{2-y}\{1 - xy^2\}$
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(29) The conditional PDF of Y given X , $f_{Y|X}(y|x)$ is

(A) $\frac{3}{3-x^2}\{1 - x^2y\}$	(B) $\frac{3}{3-x^2}\{1 - xy^2\}$	(C) $\frac{2}{2-x^2}\{1 - xy^2\}$	(D) $\frac{2}{2-x^2}\{1 - x^2y\}$
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Question (11):

A random process $X(t)$ is given by $X(t) = A$

where A is a random variable with a finite mean of μ_A and finite variance σ_A^2 .

(30) The ensemble average of $X(t)$ is

(A) $\sigma_A^2 + \mu_A^2$	(B) A	(C) μ_A	(D) σ_A^2
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(31) The time average of $X(t)$ is

(A) μ_A	(B) A	(C) $\sigma_A^2 + \mu_A^2$	(D) σ_A^2
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(32) Is $X(t)$ a mean-ergodic process?

(A) No	(B) Yes
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Question (12):

Suppose that $X(t)$ is a wide-sense stationary Gaussian process with the autocorrelation function $R_{XX}(\tau) = 6e^{-|\tau|/2}$

- (33) The expected value of $X(t)$ is
- | | | | |
|-------|-------------|-------------|-------------|
| (A) 0 | (B) ± 2 | (C) ± 3 | (D) ± 1 |
|-------|-------------|-------------|-------------|

Concerning the covariance matrix of the random variables $X(t)$, $X(t+1)$, $X(t+2)$ and $X(t+3)$,

- (34) The element C_{11} is
- | | | | |
|-------|-------|-------|-------|
| (A) 4 | (B) 6 | (C) 2 | (D) 3 |
|-------|-------|-------|-------|

- (35) The element C_{14} is
- | | | | |
|---------------|---------------|-----------------|---------------|
| (A) $6e^{-1}$ | (B) $6e^{-2}$ | (C) $6e^{-1.5}$ | (D) $6e^{-3}$ |
|---------------|---------------|-----------------|---------------|

- (36) The element C_{23} is
- | | | | |
|-----------------|-----------------|---------------|---------------|
| (A) $6e^{-0.5}$ | (B) $6e^{-1.5}$ | (C) $6e^{-3}$ | (D) $6e^{-2}$ |
|-----------------|-----------------|---------------|---------------|

- (37) The element C_{42} is
- | | | | |
|-----------------|---------------|---------------|---------------|
| (A) $6e^{-0.5}$ | (B) $6e^{-3}$ | (C) $6e^{-2}$ | (D) $6e^{-1}$ |
|-----------------|---------------|---------------|---------------|

Question (13):

- (38) In general, the transition probability matrix is

(A) stochastic	(B) symmetric	(C) doubly stochastic	(D) none of these
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Question (14):

A random process $X(t)$ has the power spectral density $S_{XX}(\omega) = \frac{9}{\omega^2 + 64}$

(Hint: consider the Fourier Transform Pair $e^{-a|\tau|} \leftrightarrow \frac{2a}{\omega^2 + a^2}$)

- (39) The autocorrelation function is

(A) $0.7525 e^{-a \tau }$	(B) $0.2556 e^{-a \tau }$	(C) $0.6252 e^{-a \tau }$	(D) $0.5625 e^{-a \tau }$
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- (40) The average power in the process is

(A) 0.2556	(B) 0.5625	(C) 0.6252	(D) 0.7525
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"GOOD LUCK"